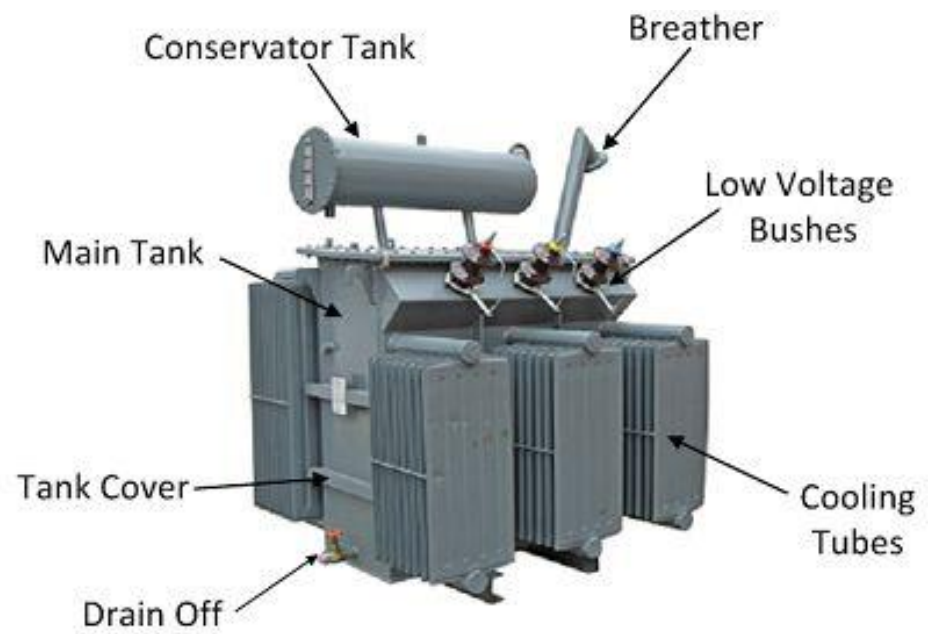
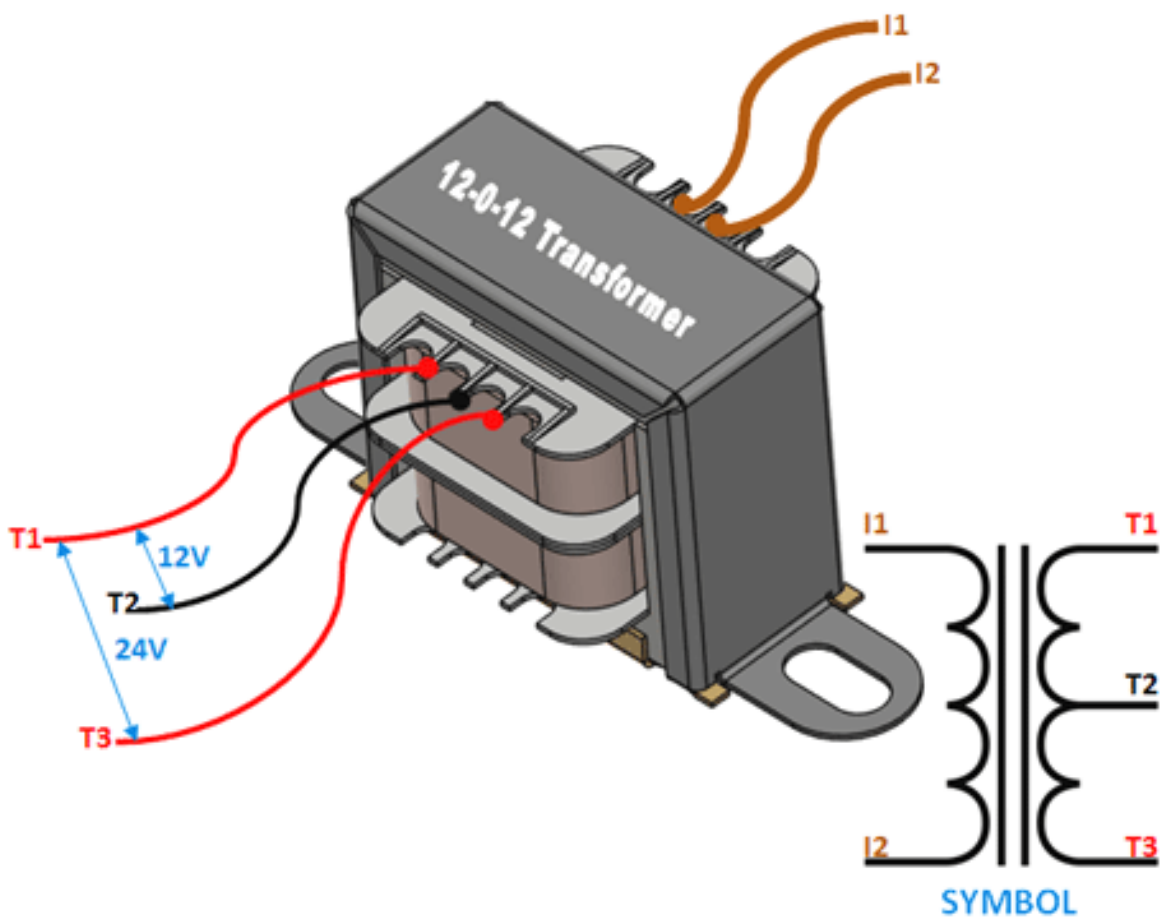


Module 5

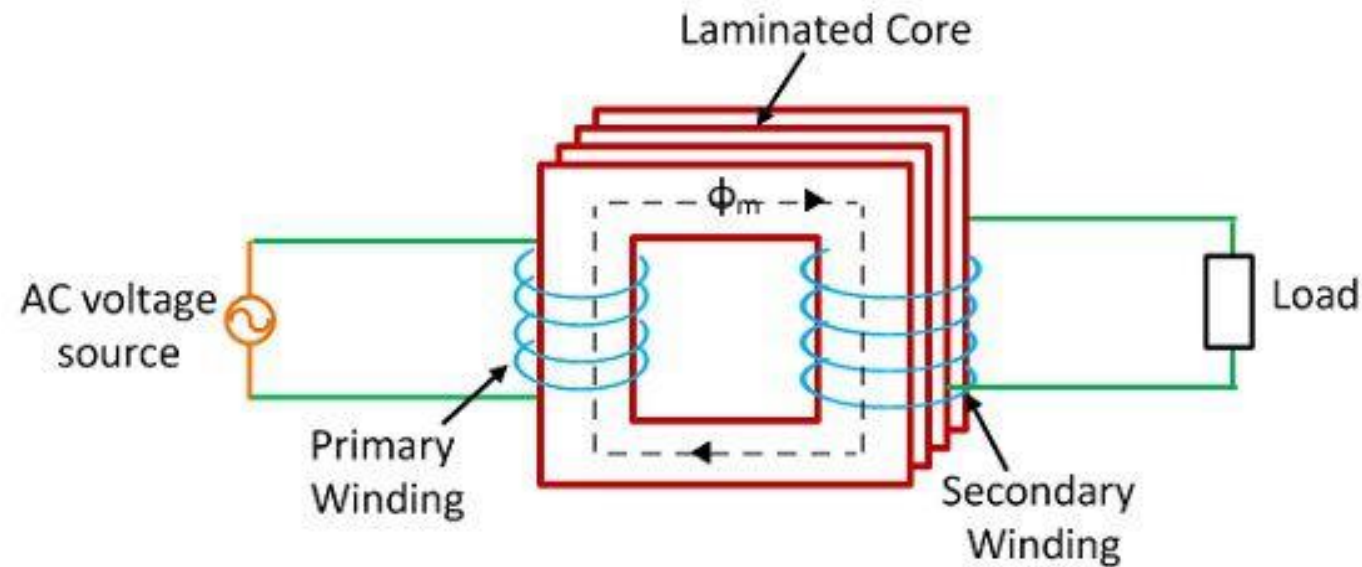
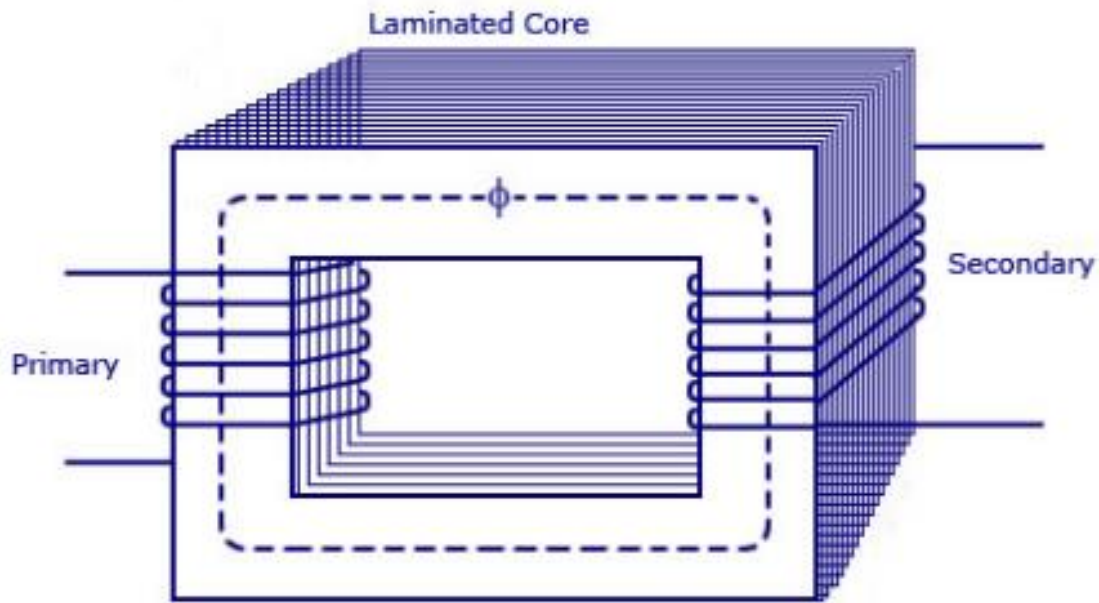
Electrical Machines





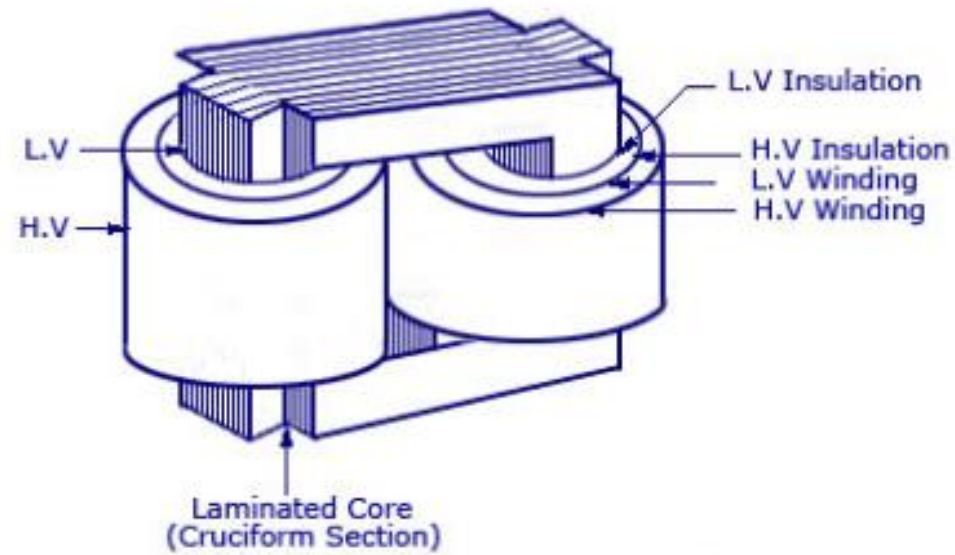
Principle of Operation

A transformer can be defined as a static device which helps in the transformation of electric power in one circuit to electric power of the same frequency in another circuit. The voltage can be raised or lowered in a circuit, but with a proportional increase or decrease in the current ratings. In this article we will be learning about Transformer basics and working principle

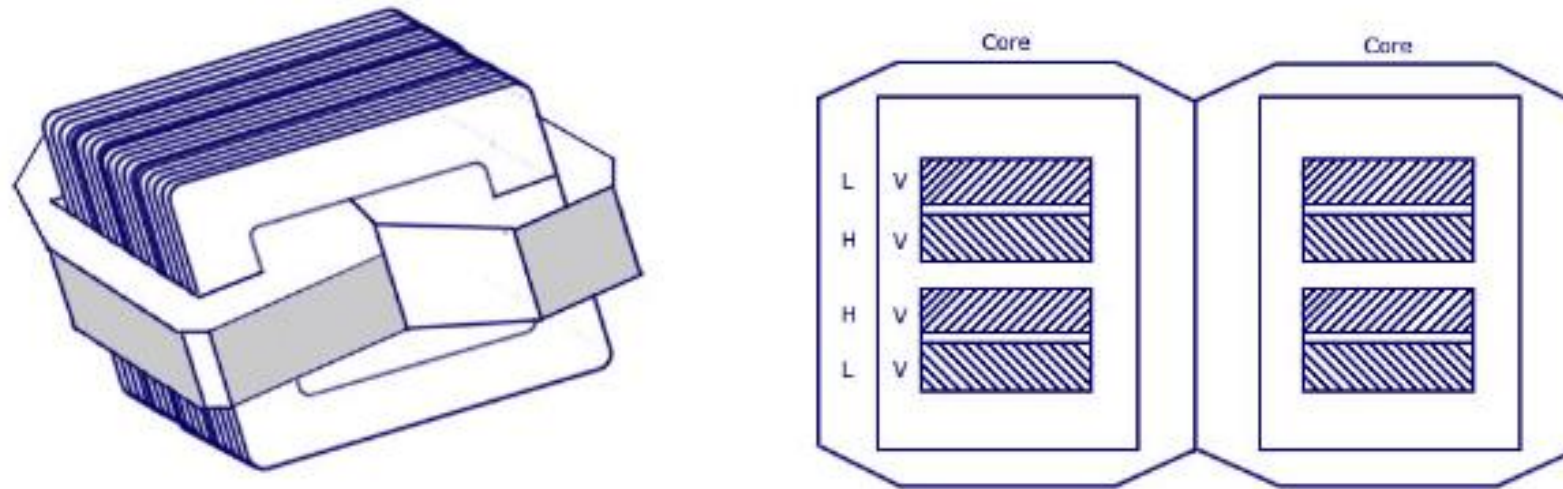


The main principle of operation of a transformer is mutual inductance between two circuits which is linked by a common magnetic flux. A basic transformer consists of two coils that are electrically separate and inductive, but are magnetically linked through a path of reluctance.

Core Type Transformer Cruciform Section

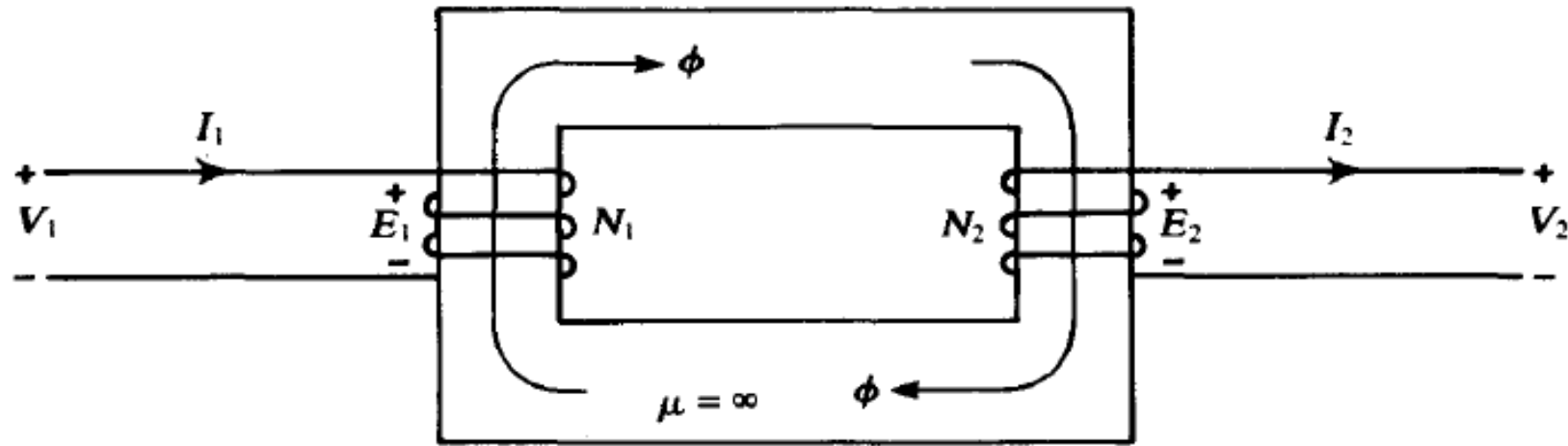


Shell Type Transformers Rectangular Form



EMF Equation

A *transformer* is an electromagnetic device having two or more mutually coupled windings.



according to *Faraday's law* of electromagnetic induction, emfs e_1 and e_2 are induced in N_1 and N_2 , given by

$$e_1 = N_1 \frac{d\phi}{dt} \quad e_2 = N_2 \frac{d\phi}{dt} \quad (1)$$

$$\frac{E_1}{E_2} = \frac{N_1}{N_2} \equiv a \quad \text{where } a \text{ is the } \textit{turns ratio}.$$

The transformer being ideal, $e_1 = v_1$ and $e_2 = v_2$ in Fig. ; hence the flux and voltage are related by

$$\phi = \frac{1}{N_1} \int v_1 dt = \frac{1}{N_2} \int v_2 dt$$

If the flux varies sinusoidally, $\phi = \phi_m \sin \omega t$, then the corresponding induced voltage, e , in an N -turn winding is given by

$$e = \omega N \phi_m \cos \omega t$$

and the rms value of this induced voltage is

$$E = \frac{\omega N \phi_m}{\sqrt{2}} = 4.44 f N \phi_m$$

Power Losses in transformer

Basically two types of losses

1. Core Loss (or) Iron Loss
2. Copper Loss

Transformer is a static device, hence mechanical losses (like friction loss) are absent in it. A transformer only consists of electrical losses (iron losses and copper losses).

Core Loss Or Iron Loss :

Core losses are due to the magnetic properties of the material used for the construction of core. Core loss is further classified into two types

1. Hysteresis loss
2. Eddy current loss

Hysteresis loss:

Hysteresis loss is due to reversal of magnetization in the transformer core.

As we know current flows in alternate direction in case of A.C. so for each cycle, the transformer core magnetizes and demagnetizes alternatively, but at a rate of 50/60 Hz there will be a lag in between magnetic flux and magnetic field, and this will leave some residual magnetic flux in the core even the magnetic field is cut off and we need some more energy to remove the residual flux, this energy loss is known as Hysteresis loss.

This loss depends upon the volume and grade of the iron(Core), frequency of magnetic reversals and value of flux density.

Eddy current loss:

In transformer, primary winding sets up an alternating magnetizing flux and is linked with the secondary winding and it induces an emf which is in fact the output, but some part of the flux gets linked with the body of the transformer as well and this circulates current in the body which is known as Eddy Current. This current produces a loss in the core which is known as Eddy Current loss.

Copper Loss In Transformer:

As pure inductive coil is not possible, it must have some resistance, so due to resistances there are losses in each winding of transformer which are called copper loss. This loss varies as load current varies so it is called variable loss.

Efficiency Of Transformer

$$\text{efficiency} = 1 - \frac{\text{losses}}{\text{input}} = 1 - \frac{I_1^2 R_1 + W_i}{V_1 I_1 \cos \Phi_1}$$

$$\eta = 1 - \frac{I_1 R_1}{V_1 \cos \Phi_1} - \frac{W_i}{V_1 I_1 \cos \Phi_1}$$

differentiating above equation with respect to I_1

$$\frac{d\eta}{dI_1} = 0 - \frac{R_1}{V_1 \cos \Phi_1} + \frac{W_i}{V_1 I_1^2 \cos \Phi_1}$$

$$\eta \text{ will be maximum at } \frac{d\eta}{dI_1} = 0$$

Hence efficiency η will be maximum at

$$\frac{R_1}{V_1 \cos \Phi_1} = \frac{W_i}{V_1 I_1^2 \cos \Phi_1}$$

$$\frac{I_1^2 R_1}{V_1 I_1^2 \cos \Phi_1} = \frac{W_i}{V_1 I_1^2 \cos \Phi_1}$$

$$I_1^2 R_1 = W_i$$

Hence, **efficiency of a transformer** will be maximum when copper loss and iron losses are equal.

Copper loss = Iron loss.

All Day Efficiency Of Transformer

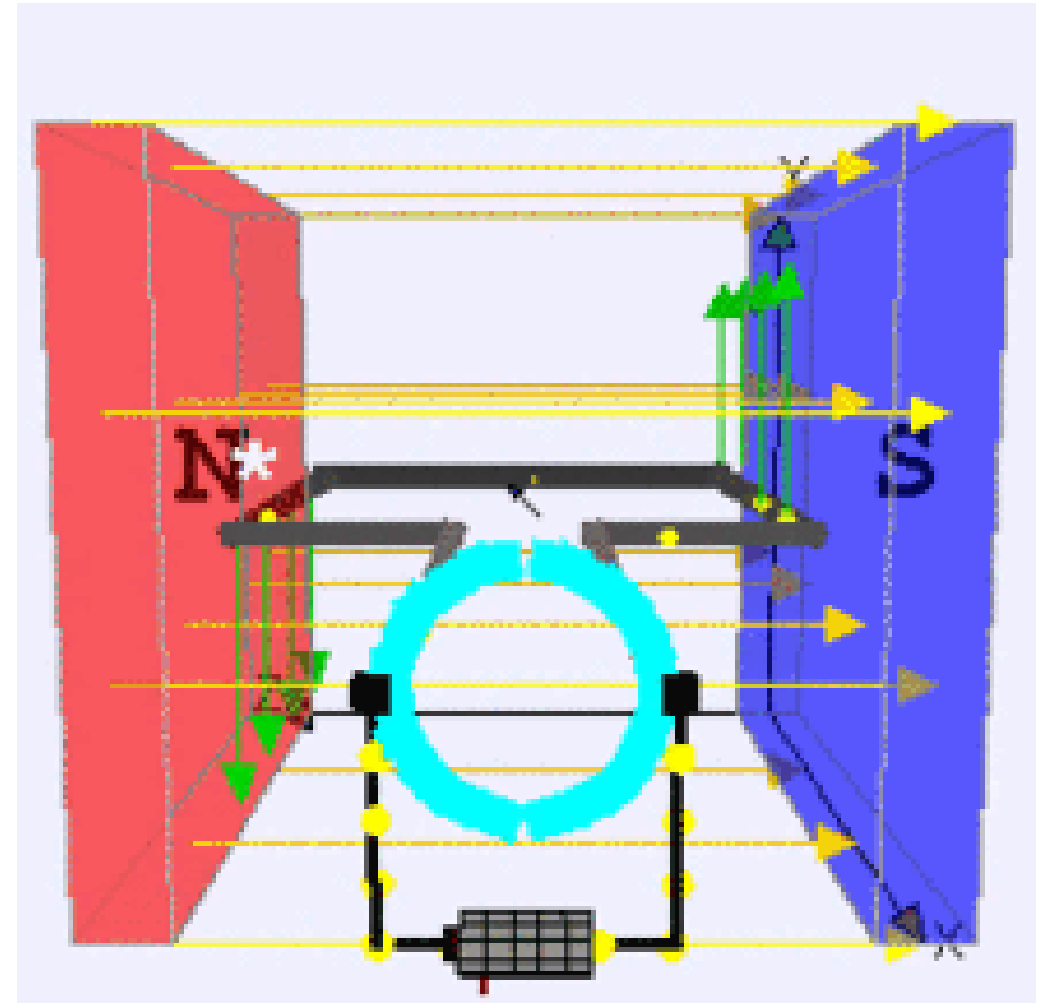
$$\text{ordinary efficiency} = \frac{\text{output (in watts)}}{\text{input (in watts)}}$$

$$\text{All day efficiency} = \frac{\text{output (in kWh)}}{\text{input (in kWh)}} \quad (\text{for 24 hours})$$

DC Machine

A DC Machine is an electro-mechanical energy conversion device. There are two types of DC machines; one is DC generator, and another one is known as DC motor.

It work from a d.c. source and generate mechanical power or convert mechanical power into a d.c. power.



Working Principle of A DC Motor

An electric motor is an electrical machine which converts electrical energy into mechanical energy.

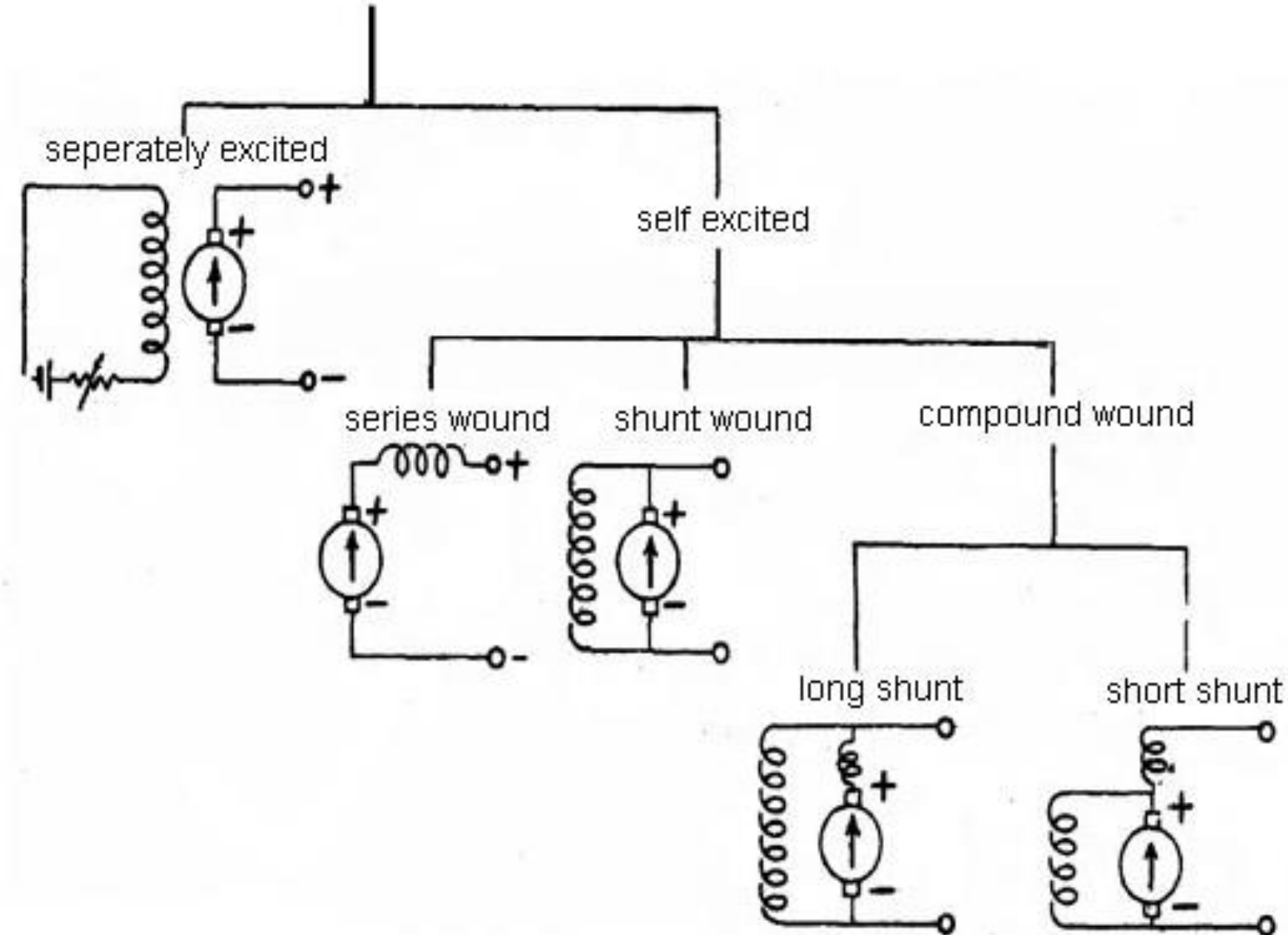
The basic working principle of a DC motor is: "whenever a current carrying conductor is placed in a magnetic field, it experiences a mechanical force".

The direction of this force is given by Fleming's left-hand rule and its magnitude is given by

$$F = BIL$$

Where, B = magnetic flux density, I = current and L = length of the conductor within the magnetic field.

Classification of DC machines



Each DC machine can act as a generator or a motor. Hence, this classification is valid for both: DC generators and DC motors. DC machines are usually classified on the basis of their field excitation method. This makes two broad categories of dc machines; (i) Separately excited and (ii) Self-excited.

Separately excited DC machines: In separately excited dc machines, the field winding is supplied from a separate power source. That means the field winding is electrically separated from the armature circuit.

Separately excited DC generators are not commonly used because they are relatively expensive due to the requirement of an additional power source or circuitry.

Self-excited DC machines: In this type, field winding and armature winding are interconnected in various ways to achieve a wide range of performance characteristics (for example, field winding in series or parallel with the armature winding).

In a **self-excited type of DC generator**, the field winding is energized by the current produced by themselves.

Self-excited machines can be further classified as –

Series wound dc machines – In this type, field winding is connected in series with the armature winding. Therefore, the field winding carries whole of the load current (armature current). That is why series winding is designed with few turns of thick wire and the resistance is kept very low (about 0.5 Ohm).

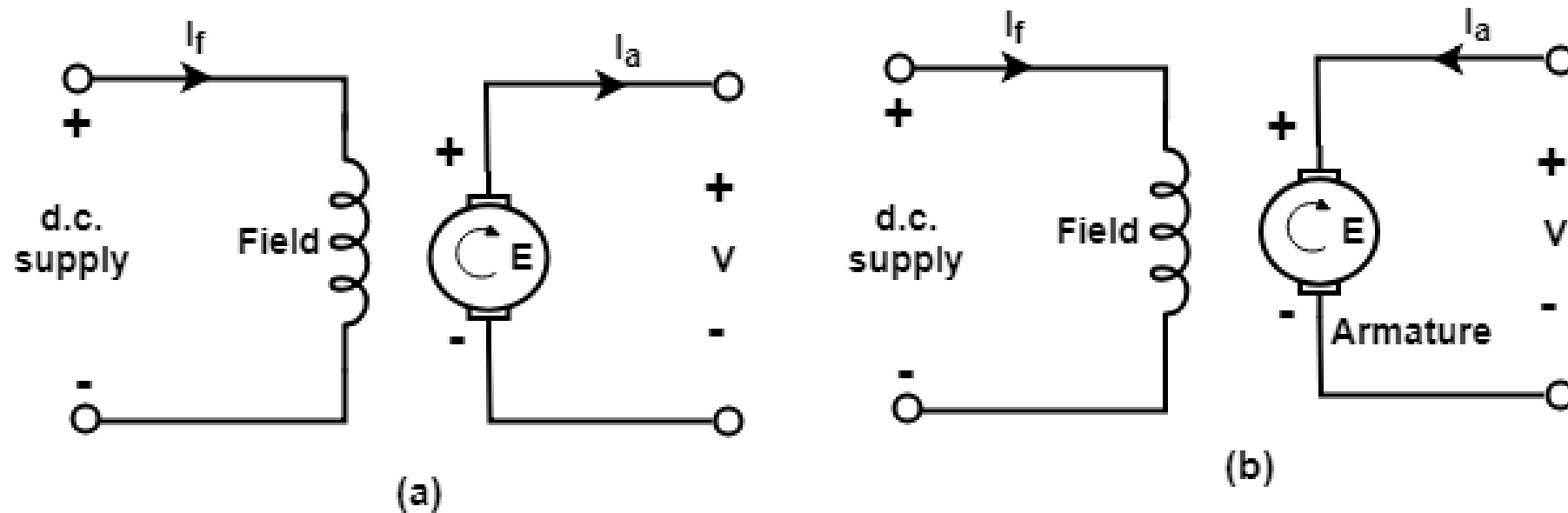
Shunt wound dc machines – Here, field winding is connected in parallel with the armature winding. Hence, the full voltage is applied across the field winding. Shunt winding is made with a large number of turns and the resistance is kept very high (about 100 Ohm). It takes only small current which is less than 5% of the rated armature current.

Compound wound dc machines – In this type, there are two sets of field winding. One is connected in series and the other is connected in parallel with the armature winding. Compound wound machines are further divided as -

- **Short shunt** – field winding is connected in parallel with only the armature winding
- **Long shunt** – field winding is connected in parallel with the combination of series field winding and armature winding

1. Separately excited D.C. machine:

When a separate D.C. source is used to energize the field coils it is called as separately excited D.C. machine. The connections showing the separately excited D.C. Machines are given in the figure.

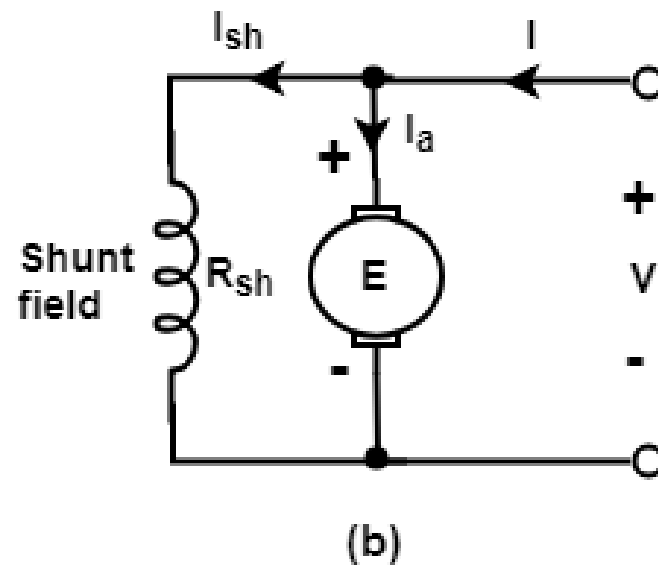
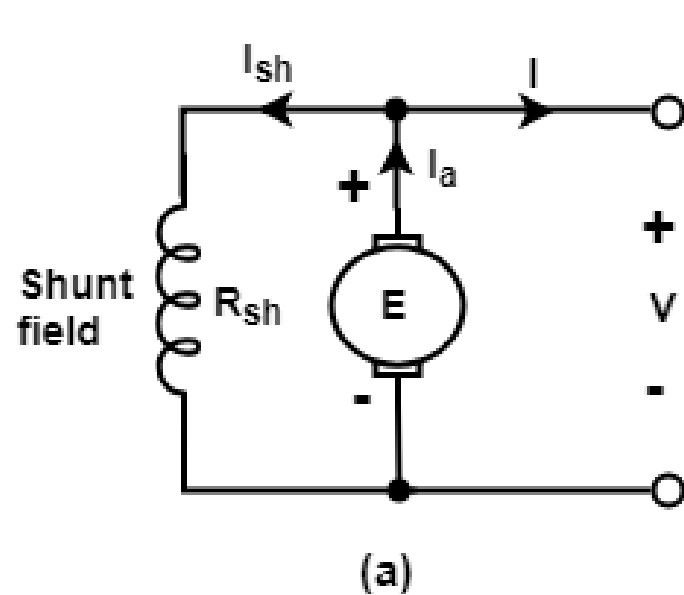


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Figure: (a) Separately excited D.C. Generator, (b) Separately excited D.C. Motor.

2. Shunt wound D.C. Machine:

Shunt wound D.C. Machines is the machine in which field coils are connected in parallel with the armature. Since the shunt field receives the full output voltage of a generator or the supply voltage of a motor, it is generally made of a large number of turns of fine wire carrying a small field current.



3. Series wound D.C. Machine:

Series wound D.C. Machines is the machine in which the field coils are connected in series with the armature. The series field winding carries the armature current, and the armature current is large, that is why series field winding consists of few turns of wire of large cross-sectional area.

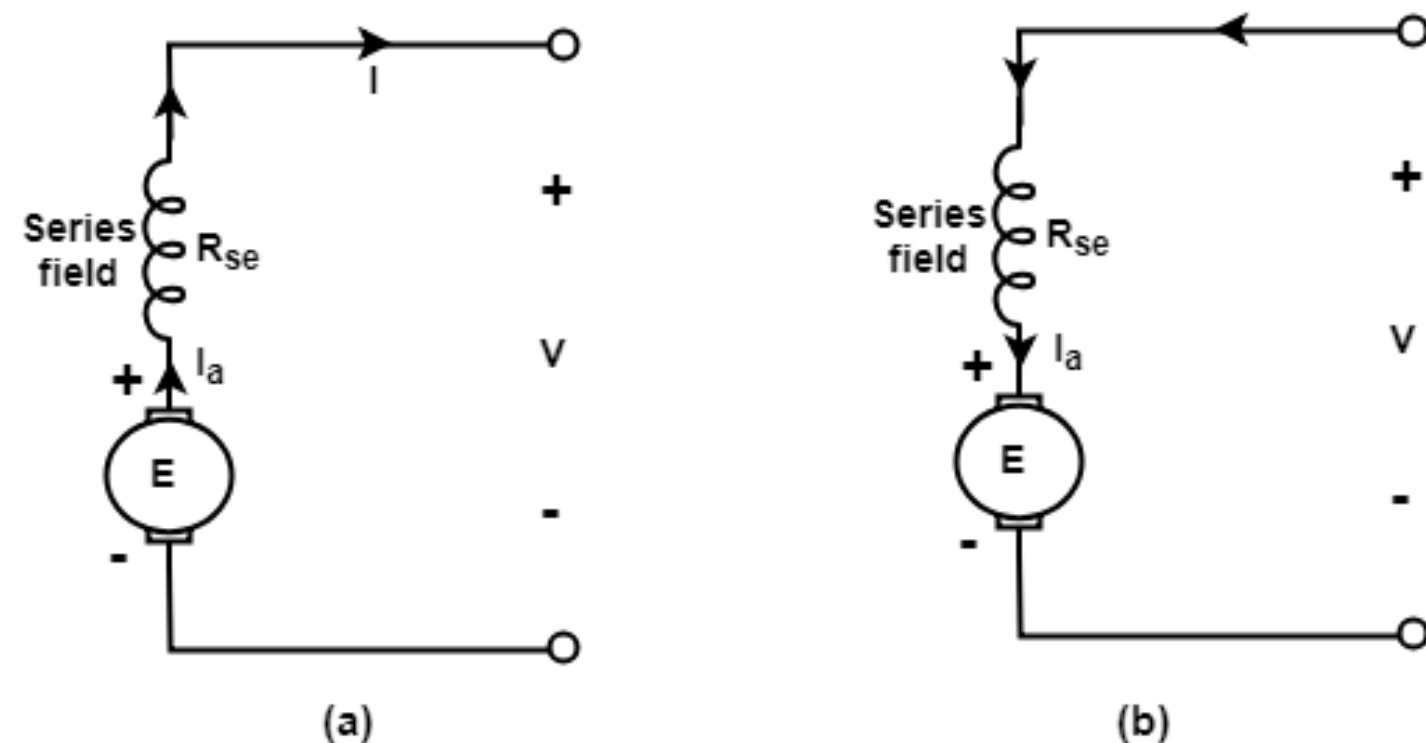
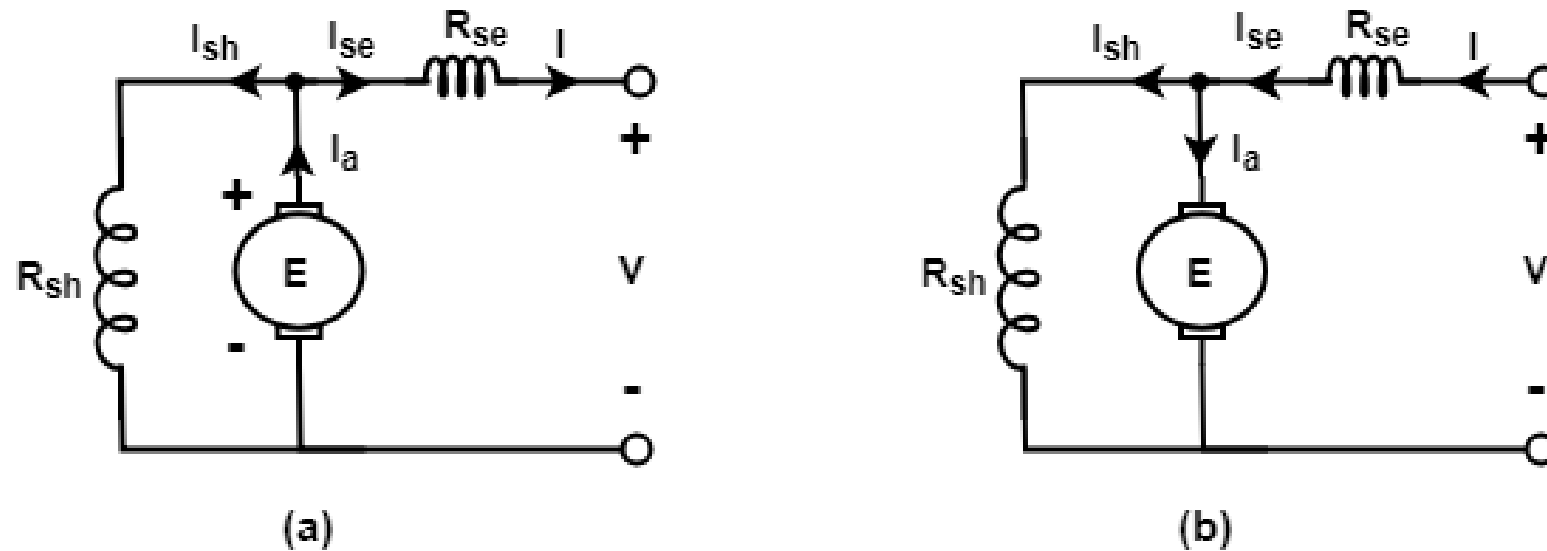


Figure: (a) D.C. series generator (b) D.C. series motor.

4. Compound wound D.C. machine:

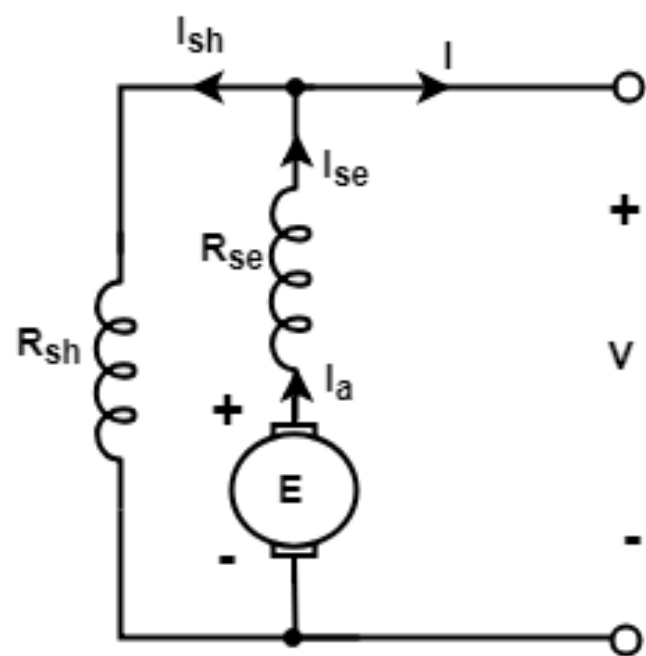
A Compound machine is a machine which has both shunt and series fields. Two windings are carried out by each pole of the machine. The series winding has few turns of large cross-sectional area, and the shunt windings have many turns of fine wire.

It can be connected in two ways. If the shunt field is connected in parallel with the armature alone, the machine is called the **short-shunt compound machine** and if the shunt field is in parallel with both the armature and series field, the machine is called the **long-shunt compound machine**.

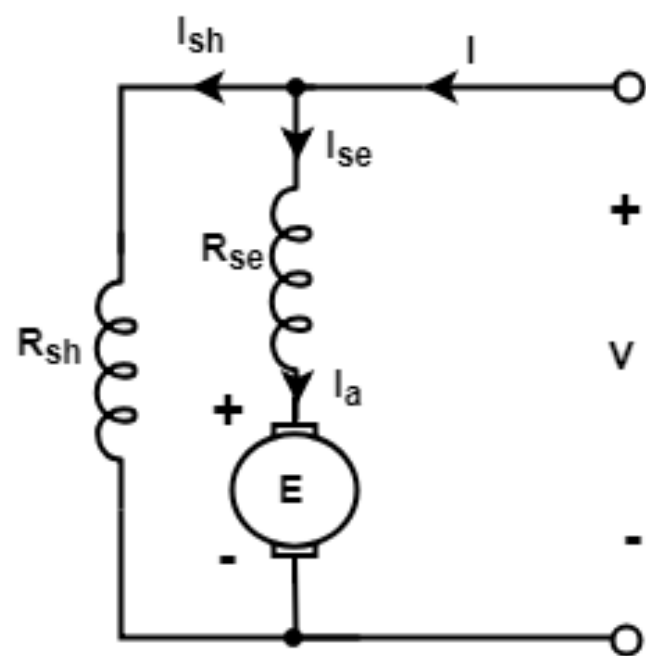


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Figure: (a) Short-shunt compound D.C. generator (b) Short-shunt compound D.C. motor.



(a)



(b)

Figure: (a) Long-shunt D.C. generator (b) Long-shunt D.C. motor.

Characteristics and Applications of DC motors

1. Separately Excited DC Motors

Characteristics of Separately Excited DC Motors

- Separately excited Dc motors got very accurate speed
- These type of motors are much suitable for applications which need speed variation from low speed to higher speed.

Applications of Separately Excited DC Motors

- Paper Machines
- Rolling Units
- Electric Populations of Ships

2. Shunt DC Motors

Characteristic of Shunt DC Motors

- Shunt motors are normally got constant speed.
- The starting torque of this motor is medium which is limited to nearly 200% by commutation.
- Speed control of shunt motor done by decreasing armature voltage control

Applications of Shunt DC Motors

Shunt Motors are normally used for constant speed applications such as followings

- Lathe Machines
- Fans and Blowers
- Printing machines
- Wood working machines
- Centrifugal pumps

3. DC Series Motors

Characteristics of DC Series Motors

- Unlike DC Shunt Motors this DC Series motors got variable speed. and the starting torque of this dc motor is quite high. Normally it can high up to 500%.
- speed regulation is widely variable at no load.
- This DC series motor speed control done by using series resistance method.
- In Dc series motors its essential to connect the load positively.

Applications of DC Series Motors

DC series motors are much suitable for applications which require high starting torque and variable speed applications. Following are some several applications of DC series motors.

- Electrical Cranes
- Trolley Cars
- Conveyors Belt Drives
- Electric locomotives

4. Compound DC Motors

Characteristic of Cumulative compound wound motors

- Normally these type of DC motor is good for adjustable varying speed.
- The speed regulation is varying 25%-30%.
- The starting torque of this dc motor is also high like DC series Motors.

Characteristic of Differential compound wound motors

- Torque and Speed of this Differential compound wound motors are almost constant.
- This motor also got the tendency toward speed instability with a possible motor running away.

Application of Compound DC Motors

DC compound motors are much suitable for places where need higher starting torque on the constant speed. The applications of DC Compound motors are as following.

- Electrical Elevators
- Conveyor Belt Drive
- Heavy Planes
- Rolling Mills
- Punches
- Shears